

Analytical Derivations and Algorithmic Details

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A. Introduction

This technical report provides supplementary materials for the paper entitled 'Spatio-Temporal Graph Reinforcement Learning for Channel Resource Allocation in C-V2X Networks'. Due to page limitations, several technical details are omitted from the main manuscript, including analytical derivations, algorithm implementation procedures, and simulation configurations. This report presents these materials in detail to provide theoretical support for the proposed analytical framework and the spatio-temporal multi-agent reinforcement learning method.

B. Notation Definition

To facilitate the presentation of the analytical models and the proposed learning framework, the major mathematical symbols used throughout this report are summarized in Table A. These notations are consistent with those adopted in the main manuscript and are frequently used in the derivations of packet reception ratio, delay analysis, and the proposed multi-agent reinforcement learning algorithm.

C. Abbreviations

Since the proposed framework involves analytical modeling, vehicular communication protocols, graph neural networks, and reinforcement learning techniques, a number of abbreviations are used throughout the manuscript. For clarity and readability, the full names of these abbreviations are summarized in Table B.

TABLE B
ABBREVIATIONS

Abbreviation	Full name
ITS	Intelligent Transportation System
DSRC	Dedicated Short Range Communications
C-V2X	Cellular-Vehicular-to-Everything
3GPP	Third Generation Partnership Project
V2V	Vehicle-to-Vehicle
V2I	Vehicle-to-Infrastructure
V2P	Vehicle-to-Pedestrian
V2N	Vehicle-to-Network
SPS	Semi-Persistent Scheduling
MAC	Media Access Control
CAM	Cooperative Awareness Message
DENM	Decentralized Environmental Notification Message
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
GCN	Graph Convolutional Network
LSTM	Long Short-Term Memory
PRR	Packet Reception Ratio
SCI	Sidelink Control Information
RC	Resource Counter
SNR	Signal-to-Noise Ratio
RRI	Resource Reservation Interval
RSSI	Received Signal Strength Indicator
MLP	Multilayer Perceptron

D. Related work on analytical models.

Existing studies have investigated the MAC layer performance of C-V2X Mode 4. Manuel et al. [1] present a packet delivery ratio model to evaluate the probability of successful packet reception. It quantifies four transmission errors, i.e., half-duplex, sensing, propagation, and packet collision ones. Zhang et al. [2] develop a packet passing rate model to assess the reliability of SPS in MAC layer communications. However, when analyzing packet-collision errors, their model neglects the interference effects caused by concurrent resource selection among multiple vehicles, which significantly impacts vehicular communication performance in dense traffic situations. In [3], a channel access collision model is proposed to describe packet collision probability. It incorporates the effects of both sensed and hidden terminals. This work [3] exclusively analyzes the transmission of periodic CAMs, but neglects DENMs. In [4], a discrete-time Markov chain model is established to describe the channel access process for CAMs and DENMs. The model prioritizes DENM transmissions, allowing CAM transmission only when the DENM queue is depleted. Employing stochastic geometry theory, Claudia et al. [5] establish a DENM loss model to characterize DENM delivery probability. Additionally, they introduce a novel reliability-latency metric to evaluate the probability of successful DENM delivery within a specified deadline. Wijesiri et al. [6] develop a multi-dimensional Markov chain model to analyze the channel access mechanism for CAM and DENM transmissions. Their work quantitatively evaluates three key performance metrics: average transmission delay, packet collision probability, and channel utilization efficiency.

For the mixed CAM and DENM environment, existing methods [4][5][6] deliver DENMs by occupying resources reserved for CAMs. For this resource sharing mechanism, the latency for DENMs is consequently constrained by the CAM resource reservation period. Meanwhile, DENM transmission disrupts the periodic schedule of CAMs. Existing models fail to model the interdependencies arising from concurrent CAM and DENM channel access. To overcome this limitation, we establish a two-dimensional Markov chain to describe the interdependent resource selection and reservation processes for both message types in a CAM-DENM coexistence scenario. Unlike existing analytical models [4] [6], which use the resource counter as the sole mechanism for resource reservation for both CAM and DENM messages, our model introduces a dedicated retransmission counter for DENM resource reservation. This is particularly important in C-V2X networks, where DENM transmissions are repeated at fixed intervals to achieve reliable reception. Hence, the proposed model explicitly characterizes how periodic CAM broadcasts and DENM repetitions interact during shared channel access.

TABLE A
NOTATION DEFINITION

Notation	Definition
$\mathbb{N}$	Set of vehicles
p_c	Probability of generating a CAM packet
p_d	Probability of generating a DENM packet
p_a	Success probability of subchannel allocation under SPS
Γ	Maximum allowable latency for CAM transmission
σ	Maximum allowable latency for DENM transmission
$\hat{b}$	Upper of RC value
$\tilde{b}$	Lower of RC value
p_k	Resource reservation probability
λ_c	Average arrival rate of CAM packets
λ_d	Average arrival rate of DENM packets
s	Number of DENM transmissions
N_{sf}	Total number of subframes within a selection window
ρ	Vehicle density
l	Number of lanes
$\tilde{d}_i$	Vehicle i 's communication range
$N_{d,i}$	Number of vehicles within vehicle i 's communication range
P_o	Sensing power threshold
P_{tr}	Transmission power
$d_{i,j}$	Distance between transmitter i and receiver j
$P_L(d_{i,j})$	Distance-dependent path loss
S_H	Log-normal shadow fading
N_0	Additive white gaussian noise
$\hat{\sigma}$	Shadow fading standard deviation
P_1	Demodulation threshold
d_{int}	Maximum interference range
V_i	Set of vehicles within vehicle i 's sensing range
V_k	Set of vehicles within vehicle k 's sensing range
V_{ik}	Set of vehicles within the common sensing range of i and k
N_{ch}	Total number of subchannels within a subframe
$\tilde{N}$	Total number of resource candidates for a vehicle
$d_{i,k}$	Distance between transmitter i and interfering vehicle k
n_i	Resource reselection time of vehicle i
n_k	Resource reselection time of vehicle k
$s_{i,k}$	Time difference between the initiation of resource selection for i and k
I_{RR}	Resource reservation interval
T_{cgI}	CAM packet generation interval
T_{dgI}	DENM packet generation interval
L_C	Packet size of CAMs
L_D	Packet size of DENMs
R_{trans}	Transmission rate of a packet
T_{max}	Maximum tolerable latency threshold
M	Set of subchannel resources
$\rho_{i,m}^t$	Binary resource selection indicator
o_i^t	Observation of agent i
a_i^t	Action space of agent i
β	Positive reward value
r^t	Reward function at t
$\mathbb{N}_t(i)$	Set of agents within the communication range of i
$\mathbb{N}_t(k)$	Set of agents within the communication range of k
N_m^t	Number of agents that select subchannel m at time slot t
$N_t(m)$	Set of agents using subchannel resource m at t
ω_0	Resource conflict penalty weight
ω_1	PRR reward weight
ω_2	Latency reward weight
$\hat{e}(i, k)$	Edge between nodes i and k
w_i	Weight factor in observation embedding layer
b_i	Bias vector in observation embedding layer
W_1^1	Trainable weight in the first convolution layer
W_2^2	Trainable weight in the second convolution layer
$\mathbb{N}_i^t$	Neighborhood set of agent i at t
γ	Discount factor that balances current and future rewards
$Q_{\theta-}$	Target Q network
Q_{θ}	Training Q network
λ	Learning rate

TABLE C
RELATIONSHIP BETWEEN Γ AND RESOURCE COUNTER

Γ	Resource counter range
100ms	[5,15]
50ms	[10,30]
20ms	[25,75]

E. The relationship between resource counter and Γ

To ensure the correct operation of the Semi-Persistent Scheduling (SPS) protocol, the value of the Resource Counter (RC) must be appropriately configured according to the maximum allowable latency of Cooperative Awareness Message (CAM) transmissions, denoted as Γ . In the standard, Γ can take values of 100 ms, 50 ms, or 20 ms, and each corresponds to a specific range of RC values from which a vehicle randomly selects when reserving a resource. The detailed mapping between Γ and the RC value range is summarized in Table C [7]. This mapping is critical in the proposed Markov chain model, as it directly determines the resource reservation period and influences the transition probabilities between different reservation states, thereby affecting the overall channel access performance under coexisting CAM and DENM traffic.

F. The first three types of errors

The first one is half-duplex error, which is usually caused by more than one vehicle transmitting packets in the same subframe. The second one is sensing error. When a received signal power is below a sensing power threshold, the sensing error occurs. The third one is propagation error that arises when the received signal-to-noise ratio is not sufficient for a receiver to successfully decode the received packets. The last one is the collision error, i.e., Multiple vehicles simultaneously select the same channel resource and broadcast their packets. The interference caused by other vehicles makes it impossible for the receiving vehicle to receive the packets correctly.

1) Half-Duplex Error

In C-V2X Mode 4, vehicles operate in a half-duplex mode. This error occurs when several vehicles broadcast their packets in the same subframe. The packets are then lost since vehicles cannot transmit and receive packets in the same subframe. Let vehicle i be the transmitter and vehicle j be the receiver. We analyze the probability of such error occurring based on subframe allocation conflicts, defined as ϕ_i^1 for vehicle i . It is defined as the probability that a subframe is selected by at least two vehicles, i.e.,

$$\phi_i^1 = 1 - \left(1 - \frac{1}{N_{sf}}\right)^{N_{d,i}} - \frac{N_{d,i}}{N_{sf}} \left(1 - \frac{1}{N_{sf}}\right)^{N_{d,i}-1}, \quad (S1)$$

where N_{sf} is total number of subframes within a selection window. $N_{d,i}$ represents the number of vehicles within vehicle i 's communication range:

2) Sensing Error

A sensing error occurs when the received signal power falls below a sensing threshold P_o [8]. To derive the probability of this event, we model the received power considering both

path loss and shadow fading. We denote ϕ_i^2 as the average sensing error probability of vehicle i , which is computed over all receivers within its sensing range, i.e.,

$$\phi_i^2 = \frac{1}{2\tilde{d}} \int_{-\tilde{d}}^{\tilde{d}} \phi^2(\xi) d\xi, \quad (S2)$$

where ξ denotes the distance between the transmitter and any receiver within the communication range. $\tilde{d}$ denotes vehicle communication range.

According to [2], the the probability that the received signal power drops below the sensing power threshold P_o is:

$$\phi^2(\xi) = \frac{1}{2} (1 - \text{erf}(\frac{P_{tr} - P_L(\xi) - P_o}{\sigma\sqrt{2}})), \quad (S3)$$

where P_{tr} denotes the transmission power and $P_L(\xi)$ represents the pathloss.

3) Propagation Error

Propagation error occurs when the received signal power is higher than P_o , yet the received signal-to-noise ratio $SNR(d_{i,j})$ is insufficient for correctly decoding a packet. $d_{i,j}$ is the distance between transmitter i and receiver j . We model the received Signal-to-Noise Ratio (SNR) considering path loss, shadow fading and noise [9]. We have:

$$SNR(d_{i,j}) = P_{tr} - P_L(d_{i,j}) - S_H - N_0, \quad (S4)$$

where N_0 denotes the noise power. The shadowing S_H follows a log-normal distribution. Thus, the PDF of the received SNR is:

$$f_{SNR,d_{i,j}}(q) = \frac{1}{\hat{\sigma}\sqrt{2\pi}} \exp(-(\frac{P_{tr} - P_L(d_{i,j}) - N_0 - q}{\hat{\sigma}\sqrt{2}})^2), \quad (S5)$$

where q represents the possible values of the received SNR.

The occurrence of a propagation error requires that: 1) the received signal power is higher than P_o ; 2) the received SNR is below a threshold P_1 . Hence, the probability that the propagation error occurs between transmitter i and receiver j is:

$$\phi^3(d_{i,j}) = \Pr\{P_r(d_{i,j}) \geq P_o\} \cdot \Pr\{SNR(d_{i,j}) < P_1\} \\ = \int_{P_o}^{+\infty} f_{P_r,d_{i,j}}(p) dp \cdot \int_{-\infty}^{P_1} f_{SNR,d_{i,j}}(q) dq. \quad (S6)$$

Let ϕ_i^3 denote the average propagation error probability of vehicle i , which is computed over all receivers within the sensing range, i.e.,

$$\phi_i^3 = \frac{1}{2\tilde{d}} \int_{-\tilde{d}}^{\tilde{d}} \phi^3(\xi) d\xi. \quad (S7)$$

G. Derivation of the number of common candidate resources for the fourth type of errors.

To calculate the number of occupied resources, we must determine the number of vehicles within vehicle i 's sensing range $\tilde{d}$. We have the set of vehicles in its range:

$$V_i = \{v_j | d_{i,j} \leq \tilde{d}_i\}. \quad (S8)$$

V_i is the set of vehicles within vehicle i 's sensing range $\tilde{d}$. $V_{ik} = V_i \cap V_k$ represents the set of vehicles that are located

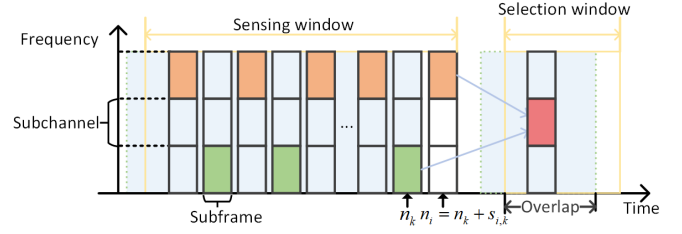


Fig. A. Resource selection conflict between two vehicles.

within the common sensing range of i and k . We define $N_{sr,i}$ as the number of resources occupied by vehicles within the sensing range of vehicle i . The number of vehicles within i 's range is $|V_i|$. The number of resource candidates for a vehicle is $\hat{N} = N_{sf} \cdot N_{ch}$, where N_{ch} denotes total number of subchannels within a subframe. The computation of $N_{sr,i}$ can be approximately modeled as distributing $|V_i|$ distinguishable balls among $\hat{N}$ distinguishable baskets and calculating the expected number of baskets that contain at least one ball [3]. For each basket, the probability that it contains at least one ball is given by $(1 - (1 - 1/\hat{N})^{|V_i|})$. Therefore, the number of occupied resources by the vehicles within i 's range is:

$$N_{sr,i} = \hat{N} (1 - (1 - \frac{1}{\hat{N}})^{|V_i|}). \quad (S9)$$

The number of resources occupied by the vehicles in V_{ik} , denoted as $N_{cr}(d_{i,k})$, is:

$$N_{cr}(d_{i,k}) = \hat{N} (1 - (1 - \frac{1}{\hat{N}})^{|V_{ik}|}). \quad (S10)$$

We define $M_R(d_{i,k})$ as the number of resources in the candidate resource set of vehicle i , i.e.,

$$M_R(d_{i,k}) = \hat{N} - N_{sr,i}. \quad (S11)$$

$|V_k - V_{ik}|$ represents the number of vehicles that are within the sensing range of k but outside the common sensing range of two vehicles. Therefore, the number of common candidate resources for i and k is:

$$M_{CR}(d_{i,k}) = M_R(d_{i,k}) [1 - \frac{1}{\hat{N} - N_{cr}(d_{i,k})}]^{|V_k - V_{ik}|}. \quad (S12)$$

When the selection windows of two vehicles overlap, they share common candidate resources. As a result, both vehicles may select the same subchannel resulting in packet collisions. Fig. A illustrates resource selection conflict between i and k . We define $s_{i,k}$ as the time difference between the initiation of resource selection for i and k , where $0 \leq s_{i,k} < I_{RR}$ that represents Resource Reservation Interval (RRI). In Fig. A, the blue area indicates the sensing and selection windows of k , with its resource reselection scheduled at time n_k . The white area represents the sensing and selection windows of i , with its reselection scheduled at $n_i = n_k + s_{i,k}$. In this case, two vehicles select the same resource for their next transmission.

As illustrated in Fig. B, the length of overlapping area between the selection windows of i and k is $I_{RR} - s_{i,k}$.

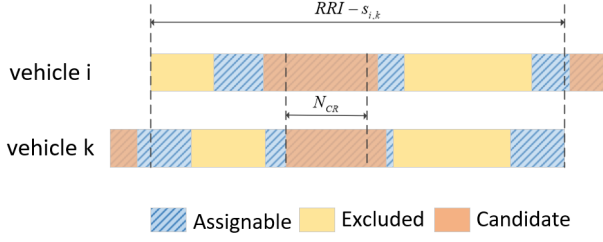


Fig. B. Excluded, assignable and common candidate resources for two vehicles.

Hence, the number of their common candidate resources, $N_{CR}(d_{i,k}, s_{i,k})$, is calculated as:

$$N_{CR}(d_{i,k}, s_{i,k}) = \frac{I_{RR} - s_{i,k}}{I_{RR}} M_R(d_{i,k}) \left[1 - \frac{1}{\hat{N} - N_{cr}(d_{i,k})} \right]^{|V_k - V_{ik}|}. \quad (S13)$$

H. Algorithm

Algorithm 1 shows a multi-agent graph reinforcement learning (MAGRL) method for subchannel resource selection in C-V2X Mode 4. The proposed MAGRL integrates GCN and LSTM to extract spatio-temporal features, respectively. These features are then used by DQN to update the policy and value functions. We detail the process starting from the filling of the experience replay buffer to the training of each agent, up until the update of the target network. When the experience buffer has enough samples, the training phase can begin. For every T_{train} , a mini-batch tuples are randomly sampled from the replay buffer to train our networks. DQN is trained through multiple episodes. During each training step, to prevent convergence to a local optimum, all agents follow an ε -greedy policy when exploring state-action space. The exploration rate ε determines the frequency of the exploration actions during learning, and is gradually reduced during training in the proposed method. The target network is updated every N_{target} time steps.

I. Simulation Parameters

To facilitate the reproduction of the simulation results reported in the main manuscript, the key simulation parameters are summarized in this section. Unless otherwise specified, all experiments are conducted using the parameter settings listed in Table D and Table E.

Algorithm 1 MAGRL for subchannel resource selection

Input: GCN structure, LSTM structure, Q-network structure, and simulation environment

Output: Resource selection policy π

- 1: Initialization: initialize the GCN, LSTM and all Q-network models. Start environment simulation and create an empty experience replay buffer $\mathcal{B}$
- 2: **for** each episode **do**
- 3: Each vehicle randomly selects an action.
- 4: **for** each step t **do**
- 5: Obtain the graph information, including node observations o^t and edge weights α^t
- 6: **for** each vehicle i **do**
- 7: Obtain feature vector x_i^t by the observation embedding;
- 8: Utilize the CCN in (31) and (32) to extract spatial feature $z_i^{t,2}$;
- 9: Utilize the LSTM in (33) to extract temporal feature h_i^t ;
- 10: Choose an action a_i^t based on state $\mu^t = (x_i^t, h_i^t)$, according to ε -greedy policy.
- 11: **end for**
- 12: Generate a reward r^t in environment simulator;
- 13: Transitions to a new state, and the spatio-temporal feature vector for the next time, $\mu^{t+1} = (x_i^{t+1}, h_i^{t+1})$, is obtained;
- 14: Store the tuple $(\mu^t, a^t, r^t, \mu^{t+1})$ in the experience replay buffer $\mathcal{B}$
- 15: **if** $T_{train} \bmod t = 0$ **then**
- 16: Randomly sample a mini-batch from the experience replay buffer $\mathcal{B}$ to train GCN with parameters W_g , and the Q network with parameters θ by minimizing the mean square error in (35);
- 17: Update the target Q-network: $\theta^- \leftarrow \theta$ every N_{target} time steps;
- 18: Update target GCN: $W_g^- \leftarrow W_g$ every N_{target} time steps.
- 19: **end if**
- 20: **end for**
- 21: **end for**

TABLE D
SIMULATION PARAMETERS

Parameter	Value
Number of vehicles N_{ve}	0...240
Highway length	2km
Maximum speed of vehicles	100km/h
Carrier frequency	5.9GHz
Channel bandwidth	10 MHz
Subchannel scheme	adjacent
Transmission power P_t	23 dBm
RBs per subchannel	10
Packet length	190 bytes
Resource reservation probability p_a	0...0.8
Resource reservation period	20ms, 50ms, 100ms
CAM packet rate λ_c	50/s, 20/s, 10/s
DENM packet rate λ_d	1/s
Noise power	-102dBm
The number of DENM retransmission s	4

TABLE E
THE PARAMETERS OF DRL MODEL

Parameter	Value
Number of training episodes	600
Initial exploration	1
Final exploration	0.01
Total exploration steps	2000
Experience-replay D	40000
Mini-batch size	512
Step size T_h for LSTM	6
Learning rate	0.001
Discount factor	0.7
Update frequency of network parameters N_{target}	200 steps
Reward weights $\omega_0, \omega_1, \omega_2$	1,0.8,0.8

REFERENCES

- [1] M. Gonzalez-Martín, et al., "Analytical Models of the Performance of C-V2X Mode 4 Vehicular Communications," in *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1155-1166, Feb. 2019.
- [2] F. Zhang, L. Shan and X. Zhao, "Mathematical Representation for Reliability of Sensing-Based Semi-Persistent Scheduling in LTE-V2X," in *IEEE Trans. Veh. Technol.*, vol. 71, no. 9, pp. 10115-10119, Sep. 2022.
- [3] X. Gu et al., "Performance Analysis on Access Collision in Semi-Persistent Scheduling of C-V2X Mode 4," in *IEEE VTC*, Norman, OK, USA, 2021, pp. 1-5.
- [4] G. P. Wijesiri N.B.A., J. Haapola and T. Samarasinghe, "A Markov Perspective on C-V2X Mode 4," in *IEEE VTC*, Honolulu, HI, USA, 2019, pp. 1-6.
- [5] C. Campolo, A. Molinaro, A. O. Berthet and A. Vinel, "On Latency and Reliability of Road Hazard Warnings Over the Cellular V2X Sidelink Interface," in *IEEE Commun. Lett.*, vol. 23, no. 11, pp. 2135-2138, Nov. 2019.
- [6] G. P. Wijesiri N. B. A, J. Haapola and T. Samarasinghe, "The Effect of Concurrent Multi-Priority Data Streams on the MAC Layer Performance of IEEE 802.11p and C-V2X Mode 4," in *IEEE Trans. Commun.*, vol. 70, no. 1, pp. 592-605, Jan. 2022.
- [7] S. Sabeeh, P. Sroka, and K. Wesolowski, "Estimation and Reservation for Autonomous Resource Selection in C-V2X Mode 4," in *PIMRC*, Istanbul, Turkey, 2019, pp. 1-6.
- [8] R. Molina-Masegosa, M. Sepulcre and J. Gozalvez, "Geo-Based Scheduling for C-V2X Network," in *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8397-8407, Sep. 2019.
- [9] J. Zang and M. Shikh-Bahaei, "Full-Duplex Multiple Access Mechanism for Connected Vehicles Operating at Different Autonomous Levels in NR eV2X VANETs," in *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 14938-14953, Sep. 2022.